1.9 COLOUR CODING OF RESISTORS

Printing values in small resistors cannot be viewed clearly. So, in small rating resistors, colour is coated on the body of resistors in circular shape. Each colour is allotted a number. This is called as colour coding.

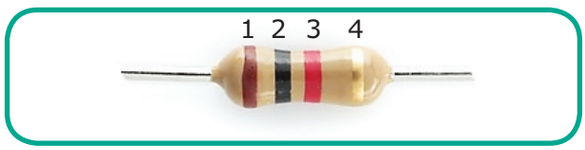


Figure 1.42(a) Color coding in a resistor

The first colour denotes the value of first digit, second colour denotes the value of second digit and the third colour (called as multiplier) denotes the number of zeros followed by the two digit. The fourth colour denotes tolerance level. If it is gold, the tolerance value of the resistor is 5% and if it is silver the tolerance is 10% of the value. Figure 1.42(a) shows the resistor with colour code. Table 1.1 shows the colour coding of the resistor.

Table 1.1 Colour Coding of Resistors

Colour	value	Multiplier	Tolerance
Black	0	10^0	-
Brown	1	10^1	±1%
Red	2	10^2	±2%
Orange	3	10^3	-
Yellow	4	10^4	-
Green	5	10^5	-
Blue	6	10^6	-
Violet	7	10^7	-
Grey	8	10^8	-
White	9	10^9	-
Gold	-	10^{-1}	±5%
Silver	-	10^{-2}	±10%
No colour	-	-	±20%

ACTIVITY 1.3

Make a colour circle showing Resistor values using colour codes as shown in the Figure 1.42(b).

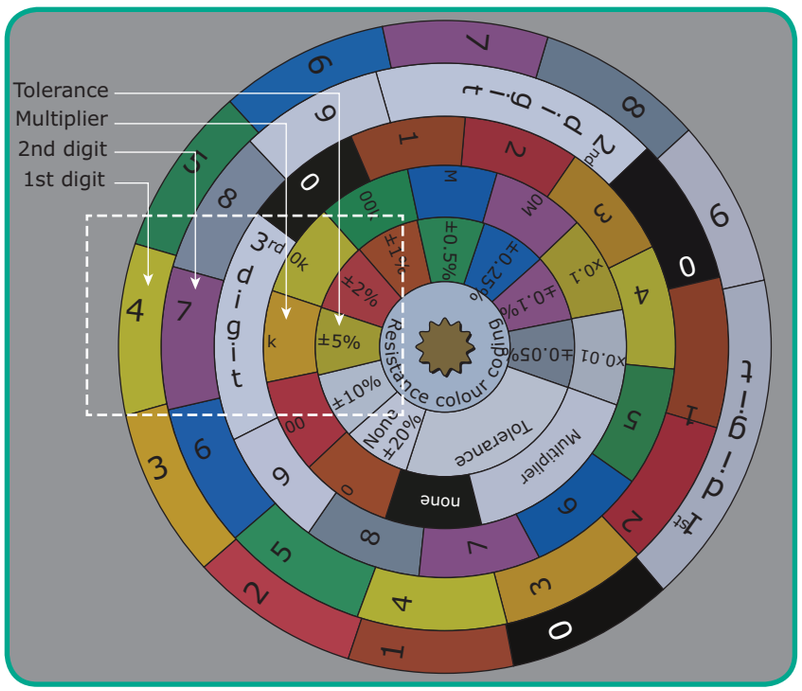


Figure 1.42(b) Colour circle for colour coding of resistors

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